

Thaneesan Siventhirarajah

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Mechatronics Engineering graduate with interdisciplinary experience in product design, additive manufacturing, computer vision, robotics, and jewellery manufacturing. Skilled in CAD modeling, rapid prototyping, 3D printing, laser technologies, and engineering R&D. Committed to applying research and Digital Fabrication expertise to develop innovative solutions for real-world engineering challenges.

Experience

Freelance Researcher

Self-employed

2024 - Present

Delivered projects in CAD design, robotics, computer vision, and additive manufacturing.

CAD Modeling and 3D Printing | Co-founder

Tensor Labs

October 2024 - September 2026

Product prototyping and additive manufacturing.

Jewellery Manufacturing Manager

Ruby Jewellery - Family Jewellery Workshop

May 2025 - Present

Jewellery design, SLA 3D printing, Casting, Laser welding, Laser engraving, and Gold refining.

Research Assistant Intern

LE Robotics (Pvt) Ltd.

June 2024 - September 2024

Worked on a computer vision project focused on warehouse applications.

STEM Workshop Instructor

2022 - Present

Conducted STEM workshops for schools and personal events, focusing on robotics, and engineering basics.

Education

BTech Hons (Mechatronics)

(September 2020 - September 2024)

Bachelor of Technology Honours in Science and Technology specialized in Mechatronics,
Faculty of Applied Sciences of Uva Wellassa University of Sri Lanka.

- GPA - 3.72 / 4.00 (First Class)
- Medium of Instruction - English
- Effective Date of the Degree - 30 September 2024

Developing Career Skills

(2018)

Department of Technical Education & Training Sri Lanka, Ministry of Skills Development & Vocational Training

G.C.E (A/L) Physical Science Stream

(2018)

Jaffna Hindu College

Z-Score - 1.2306

Combined Mathematics - B | Physics - B | Chemistry - B

G.C.E (O/L)

(2015)

Jaffna Hindu College

Mathematics - A | Science - B | English - B | ICT - A

Research Interests

- Product Design and Development
- Robotics and Automation
- Soft Robotics and Digital Fabrication
- Additive Manufacturing, Smart Manufacturing
- Computer Vision
- CAD/CAM
- Jewellery Design and Manufacturing

Projects

- Identify boxes in a warehouse (Researched on various methods YOLO, U-Net, etc.)
- Voice-Controlled Prosthetic Hand with Keyword Spotting (Python, CNN, NLP, CAD Modeling and Prototype development)
- Obstacle Detection System (YOLOv8, TensorFlow, Raspberry Pi)
- Smart Chair Walking Stick (CAD Modeling and Prototype development, ESP32 and IOT based system)
- Inclined Bucket Conveyor Belt System (CAD Modeling and Prototype development, Arduino)

Publications



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- Integrating Contemporary Casting Technologies in Traditional Goldsmithing: A Case Study on Gold Decker Ring Production: ISSN: 3021-6656
 - YOLOv8 powered solutions for box identification in warehouses
DOI: 10.38124/ijisrt/IJISRT24SEP1017
 - Voice-controlled prosthetic hand with keyword spotting: Conference: IRCUWU2024

Certifications

- Additive Manufacturing: Resin 3D Printing: LinkedIn Learning
- Rapid Prototyping for Product Design: LinkedIn Learning
- Modeling and Simulation with Simulink: Coursera
- Simulink Onramp: MathWorks
- MATLAB Onramp: MathWorks
- SOLIDWORKS CAD Design Associate (CSWA): Dassault Systèmes
- SOLIDWORKS Sheet Metal: Udemy
- SOLIDWORKS: Become a Certified Professional Today (CSWP): Udemy
- Introduction to CNC Programming: Udemy

Expertise

Computer Vision
Machining Learning
Python, C, C++

CAD Modeling
3D printing and slicing
Prototyping and Fabrication

Embedded System
Arduino, ESP 32, Raspberry pi
Jewellery manufacturing

Soft skills

Time Management
Leadership

Communication
Work under pressure

Good team player
Problem-Solving

Language

English (Professional Working)

Tamil (Native)

Sinhala (Elementary)

Honors & Awards

- Dean's list for outstanding academic performance (5 Semesters)
- LinkedEng English Camp 2021 (Participation in the Speech contest and Debate competition)
- X-mas Revamp (The most creative Category 2nd Place)

Personal Details

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- LinkedEng English Camp 2021 (Participation in the Speech contest and Debate competition)
- X-mas Revamp (The most creative Category 2nd Place)

References

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